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Rising Stars - Summer 2026 Research Internship
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Work Summary

The folder contains the modeling, simulation, and control work developed during Summer 2026 for the **Quanser Rotary Double Inverted Pendulum (DBIP)**. The purpose of this document is to explain the organization of the project folder, which contains three main sections.

1 Documentation

This folder contains the following documents:

- DBIP_ComputerSetup.docx
- Rotary Double Inverted Pendulum - Laboratory Guide.pdf
- TrackingSample.mp4
- University_of_Waterloo__Research_Report-3.pdf

The first document provides information about the computer credentials, which are stated again below:

- Username: **Administrator**
- Password: **ECE2o17b00m**

The second document is Quanser's laboratory guide for the DBIP, in which they explain the LQR + integral action controller. It is useful for getting a general understanding of the setup.

The third file is a video sample of the pendulum implemented using a structurally stable synthesis controller (Chapter 8 - Multivariable Control System ECE 488).

The fourth document is the most important one because it contains the complete mathematical derivation of the model and records the different control strategies employed to stabilize the pendulum. It is recommended to read this document first to gain a better understanding of the setup.

2 Simulation-Software

This folder contains all the relevant files required to simulate the nonlinear and linear models of the plant.

- DBIP_SymbolicDerivation.m
 - This is the first script to run. It performs the symbolic derivation of the nonlinear plant and saves the model structure to a MAT file.
- DBIP_NonlinearModel_StateFeedback.m
 - This is the second script to run. It substitutes the symbolic parameters with their numerical values and uses ode45 to solve the dynamics of the pendulum for a given initial condition and desired state using a full-state feedback controller. The results are saved to a MAT file.
- DBIP_3DAnimation.m
 - This is the third script to run. Once the results from DBIP_NonlinearModel_StateFeedback.m are saved to a MAT file, this script can be run. It provides a visual representation of the pendulum dynamics.
- DBIP_CtrlStrategies.m
 - This script allows you to choose between the different control strategies implemented on the **linear model** of the DBIP. You can modify the poles of the controllers and observers. It runs the simulation using the Simulink file called “ModularDBIPSimulation.slx”. Six different control strategy options are available.
- ModularDBIPSimulation.slx
 - Block diagram containing the plant and the different control strategies.

3 HIL_Software

This folder contains all the relevant files required to run the hardware-in-the-loop simulations.

- Jang_Peter_DBIP_setup.m
 - This script must be run first. Similar to the “DBIP_CtrlStrategies.m” script, it allows you to choose the desired control strategy. Once the control strategy has been selected, you must open the respective Simulink file.

Once the Simulink file is open, select QUARC and press Build. Once the build is complete, select QUARC again and press Start.

Monitor the alpha measurement and make sure there are no spikes in the readings. If you notice any spikes, press Stop. Then resume by selecting QUARC and pressing Start again.

Once the simulation has started, the alpha reading should be at approximately -180 degrees. Raise the pendulum counterclockwise and ensure that the encoder cable does not form any coils, as this may destabilize the pendulum for some controllers.

Once the bottom and top links are within 1.5 degrees of the upright position, the controller will take action and you must let go of the pendulum.

Once you are finished with the simulation, be prepared to catch the pendulum before pressing the Stop button.

– Results \plot_response_dbip.m

- * This script plots the results from the most recent hardware simulation, providing information about the three joint angles and the commanded voltage.



Figure 1: DBIP - Initial Condition.

Note: Raise the pendulum counterclockwise to prevent the encoder cable from forming a coil.